

Kassem Kalach

*Turning Cryptography Into a Strategic Advantage—Agile, Unified,
Quantum-ready, Simplified, and Cost-effective by 30%.*

Profile

A hands-on, operational Cryptographic Architect and Founder with 15+ years of experience building scalable, resilient cryptographic foundations across Financial Services, Industrial IoT, Critical Infrastructure, and Healthcare sectors. Specializes in transforming complex security challenges into strategic advantages—delivering agile, effective and efficient solutions that reduce complexity and cost while meeting stringent compliance and enabling high-assurance performance.

PKI & Certificate Lifecycle Management: Designing and automating unified PKI systems across hybrid and multi-cloud environments.

Post-Quantum Cryptography (PQC) Transition: Delivering crypto-agility frameworks and migration roadmaps with parallel execution paths for timely, flexible transitions.

Secrets, Key Management & Data Protection: Architecting scalable, automated lifecycles for secrets, and robust data protection across all environments, aligned with zero-trust principles.

Authentication & Identity Systems: Establishing phishing-resistant MFA, agile IAM architectures, FIDO2, and passkey strategies.

Governance, Risk & Compliance (GRC): Unifying enforceable policies aligned with NIST, PCI-DSS, eIDAS, ISO 27001, and GDPR.

Team Enablement & Knowledge Transfer: Upskilling teams via tailored training, technical mentoring, and secure-by-design enablement.

A proven leader who delivers strategic clarity, adaptable execution, and organizational alignment—to future-proof infrastructure, reduce operational overhead, and enable secure, efficient scaling.

Technical Proficiencies—Cryptography & Cybersecurity

Cryptographic Foundations	PKI & Certificate Lifecycle Management; Key & Secrets Management (HSM, KMS); Digital Signatures & KEMs (ECDSA, Dilithium, SPHINCS+, Kyber); Hashing & KDFs (SHA-2, SHA-3, HKDF, Argon2); Authenticated Encryption (AES-GCM, ChaCha20-Poly1305); Platform Firmware Resiliency: Secure Boot, & Code Signing; Key Attestation (TPM, Secure Enclave); FIDO2 / Passkeys; Crypto Agility & Post-Quantum Readiness.
Layers & Protocols	TLS 1.2/1.3; QUIC; DTLS/CoAP; SSH; Kerberos; PAKE protocols; IPsec; VPNs; S/MIME; OpenPGP; MQTT; X.509; Certificate Transparency.
Identity & Infrastructure	Identity Federation (OpenID Connect, OAuth 2.0, SAML 2.0); Single Sign-On (SSO); MFA & Phishing-Resistant Methods; Privileged Identity Management (PIM); Access Control Models (RBAC, ABAC); Network Segmentation; PKI Integration; Policy Enforcement (PEP/PDP).
Data Security & Privacy	End-to-end data protection across hybrid environments: encryption strategy (field-level, TDE, tokenization); access control & secure configuration; secrets & key management; security by design.

Cloud Security	AWS: KMS, CloudHSM, IAM, encryption (client/server), VPC; GCP: KMS, Certificate & Secret Manager; Azure: Key Vault.
Frameworks & Governance	NIST: Cybersecurity Framework (CSF), Risk Management Framework (RMF), Privacy Framework, ZTA (SP 800-207); Industry: COBIT, CSA Cloud Controls Matrix (CCM), MITRE ATT&CK, OWASP Top 10. Government of Canada: IT security risk management (ITSG-33), ITSP.40.111 (Cryptographic Algorithms),
Compliance & Standards	NIST SPs (800-30, 800-53, 800-56A, 800-57, 800-63, 800-131A, 800-171, 800-175B, 800-207, 1800-16); FIPS 140-2/3 (Validated Modules); ISO/IEC 27001 & 27017; PCI-DSS 4.0; eIDAS; GDPR; PIPEDA.
PQC Readiness	NIST PQC Standards (Dilithium, SPHINCS+, Kyber); PQC Migration Strategy (NIST/QRWG alignment); Hybrid Cryptography; Crypto-Agility Engineering; Quantum Threat Assessments; Cryptographic Inventory & Dependency Mapping;
Certifications	AWS Certified Security – Specialty (SCS-C02) (<i>Expected 2025</i>); Certified Information Security Manager (CISM) (<i>Expected March 2026</i>).

Professional Experience

Founder and CEO — Cryptimum

Sep 2024 – Present

Founded and leading a specialized consultancy delivering agile cryptographic infrastructure and strategy to Financial Services, Industrial IoT, Critical Infrastructure, and Healthcare sectors. Our mission is to transform complex security challenges into strategic advantages—reducing complexity and cost while meeting stringent compliance and enabling streamlined operations.

Core Expertise: Delivering end-to-end solutions in PKI modernization, secrets and key management architectures, post-quantum readiness, and cryptographic governance.

Value-Driven Engagement: Partnering with clients to de-risk critical environments, eliminate costly technical debt from legacy systems, and ensure enforceable compliance with evolving standards.

Thought Leadership: Continuously researching advancements in cryptography and regulatory landscapes to architect future-proof, cutting-edge solutions for organizations.

Principal Cybersecurity Architect – UKG

April 2023 – Aug 2024

Global Cryptography Owner & Strategist: Directed enterprise-wide initiatives to architect resilient cryptographic infrastructure, modernize hybrid environments, and drive compliance and operational efficiency across microservices and multi-cloud platforms. Key projects:

- ▶ Global PKI Modernization & Hybrid Cloud Integration — Principal Architect
- ▶ Enterprise Cryptographic Controls Modernization — Principal Cryptographer
- ▶ Scalable Key Management for Secure Messaging — Solutions Architect
- ▶ Enterprise PKI Inventory Tiger Team — PKI Lead
- ▶ Distributed Secrets Management for Microservices — Lead Security Architect
- ▶ Hybrid PKI Integration — Cloud Security Architect
- ▶ Secure Root CA Ceremony Design & Execution — Principal Architect

 GLOBAL PKI MODERNIZATION & HYBRID CLOUD INTEGRATION

Architected and led the redesign of a global enterprise PKI to de-risk hybrid cloud transition, strengthen regulatory compliance, and establish a scalable trust foundation. Delivered an agile trust hierarchy, multi-tier CA structure, geo-redundant intermediates, and automated certificate lifecycle management.

- **Unified Hybrid Crypto:** Engineered an agile PKI hierarchy bridging on-premises and cloud environments—eliminating siloed trust and establishing unified certificate authority across the enterprise.
- **Enabled Global Scaling:** Designed a multi-tiered CA architecture that accelerated certificate issuance by 60%—supporting a 3x increase in secure connections across NA/EMEA.
- **Improved Regulatory Resilience:** Implemented geo-redundant, region-aware intermediate CAs to meet data sovereignty requirements (e.g., GDPR) and ensure 99.99% availability.
- **Operational Efficiency:** Automated certificate lifecycle management and enforced modern cryptographic standards (TLS 1.3, ECC, AES-256)—anchored to HSM-protected root CA.

ENTERPRISE CRYPTOGRAPHIC CONTROLS MODERNIZATION

Spearheaded a comprehensive modernization of the enterprise cryptographic framework—retiring legacy algorithms, integrating advanced controls, and ensuring regulatory alignment.

- **Modernized Crypto Standards:** Led the adoption of modern algorithms—including SHA-3, AES-256, ChaCha20-Poly1305, ECDSA, and TLS 1.3—to eliminate legacy cryptography and enhance interoperability across platforms.
- **Embedded Advanced Key Management:** Architected and implemented a scalable envelope encryption strategy with strict key isolation across sensitive workloads.
- **Unified End-to-End Protection:** Established consistent cryptographic controls across data, infrastructure, and networks to meet NIST, PCI-DSS, HIPAA, and GDPR requirements.

Tech Stack AES-GCM, ECDSA, SHA-3, TLS 1.3, ChaCha20-Poly1305, Key & Secrets Lifecycle Management, Envelope Encryption, NIST, FIPS 140-2, PCI-DSS 4.0, GDPR, HIPAA.

SCALABLE KEY MANAGEMENT FOR SECURE MESSAGING

Engineered a high-performance, scalable key management framework for Kafka-based secure messaging—resolving throughput bottlenecks and compliance gaps.

- **Performance at Scale:** Implemented envelope encryption optimized for low-latency processing under KMS rate limits and high message volumes.
- **Secure Key Lifecycle:** Controlled access and lifecycle to multi-region keys via KMS and secrets management—with strict policy enforcement across services.
- **End-to-End Message Protection:** Enabled end-to-end encryption (at rest and in transit) for messaging systems—ensuring compliance with FIPS 140-2 and PCI DSS.

Tech Stack Global Data Protection, Envelope Encryption, KMS & Key Management, PCI DSS, Kafka, Pub/Sub.

ENTERPRISE PKI INVENTORY TIGER TEAM

Appointed as the PKI SME for a high-priority asset discovery initiative to reduce outage risk and establish sustainable cryptographic governance. Delivered an actionable discovery strategy, standardized inventory schema, and a foundation for long-term PKI hygiene.

- **Discovery Framework:** Developed a CA-type and environment-based discovery model to classify over 20,000 certificates using a standardized, schema-driven approach.

- **Execution & Validation:** Partnered with infrastructure teams to select discovery tools and validate records into a unified, asset-centric source of truth.
- **Governance Baseline:** Transformed legacy certificate sprawl into a structured, governed inventory aligned with enterprise crypto hygiene goals.

DISTRIBUTED SECRETS MANAGEMENT FOR MICROSERVICES

Designed a secure, cloud-agnostic secrets management solution to protect dynamic microservices and CI/CD pipelines—balancing agility with strong governance.

- **Resilient Distributed Architecture:** Engineered a multi-cloud architecture for secure secrets access across ephemeral workloads—aligned with the microservices paradigm.
- **Governance Framework:** Implemented RBAC, vault access policies, and full lifecycle auditability to enforce strong controls and visibility.
- **DevSecOps Integration:** Integrated GCP IAM and OIDC authentication into CI/CD pipelines—to eliminate hardcoded secrets and improve developer experience.

Tech Stack Cloud Secrets Manager, OpenID Connect (OIDC), Microservices, RBAC, CI/CD.

HYBRID PKI INTEGRATION FOR GCP

Integrated enterprise PKI with GCP to unify trust across hybrid environments and enforce certificate-based zero trust authentication.

- **Unified Trust Architecture:** Enabled cloud workloads to seamlessly use enterprise-issued certificates for mutual TLS across hybrid infrastructure.
- **Zero Trust Compliance:** Architected a certificate-based authentication model aligned with enterprise identity, segmentation, and zero trust policies.

SECURE ROOT CA CEREMONY DESIGN

Designed the enterprise Root CA ceremony to establish an immutable trust foundation for the full PKI infrastructure—later executed by assigned key ceremony operators.

- **Security-First Protocol:** Defined strict, security-first Root CA ceremony protocol: physical access controls, HSM-backed key generation, and defined roles (e.g., custodians, witnesses).
- **Resilient Key Management:** Established secure procedures for root key storage, encrypted backup, and disaster recovery to ensure business continuity.
- **Auditability:** Designed tamper-evident audit logs and comprehensive documentation aligned with NIST and FIPS 140-2 assurance requirements.

Tech Stack **Summary:** PKI Modernization, Hybrid Trust, Certificate Automation, OpenSSL, HSMs, TPMs, IAM, Secrets & certificate Managers (DigiCert, TLS 1.3, ECC, AES-GCM, ChaCha20-Poly1305, SHA-3, Kafka, Envelope Encryption, OIDC, RBAC, Zero Trust Architecture, GDPR, PCI DSS 4.0, NIST 800-53/57/1800-16, FIPS 140-2, HIPAA, Disaster Recovery, Cryptographic Governance, Python.

Senior Cybersecurity Researcher – Arctic Wolf

Jun 2021 – April 2023

Cloud Detection Engineering & Threat Visibility: Led research and development of proactive, platform-specific detections across AWS, Google Workspace, Microsoft, and GCP. Collaborated cross-functionally to improve monitoring, reduce attack surface, and enforce secure configurations—through automation and continuous refinement. Key projects:

- ▶ AWS Security Hub & GuardDuty Coverage — Lead Engineer
- ▶ Google Workspace Detections Roadmap — Lead Detection Engineer

- ▶ MFA Fatigue Attack Detections — Identity Security Engineer
- ▶ GCP Threat Detection Development — Cloud Security Specialist
- ▶ Microsoft Defender for Cloud Apps Ruleset — Lead Detection Engineer

📁 AWS SECURITY HUB & GUARDDUTY DETECTIONS

Designed the roadmap and implementation of detection rules across AWS Security Hub and GuardDuty—to enhance coverage of S3, EC2, access keys, and IAM-related threats.

- **Threat Visibility:** Detections for MFA misuse, access key anomalies, & API access risks.
- **Automation & Remediation:** Integrated detection outputs with automated responses to reduce Mean Time To Recover (MTTR) in AWS environments.

📁 GOOGLE WORKSPACE DETECTIONS ROADMAP

Developed and led a proactive detection roadmap for Google Workspace—targeting suspicious users, privilege escalation, and anomalous activity. Integrated alerting and consulted on detection tuning.

- **Strategic Roadmap Execution:** Defined detection priorities and success criteria for Google Workspace attack surfaces.
- **Detection Engineering:** Designed rules for suspicious logins, privilege misuse, and lateral movement attempts.
- **Security Integration:** Partnered with engineering teams to embed detections into real-time monitoring and alerting infrastructure.

📁 MFA FATIGUE ATTACK DETECTIONS

Developed detections to identify and mitigate MFA fatigue attacks in Duo, Okta, and Microsoft environments. Strengthened MFA visibility and security.

- **Adversary Technique Coverage:** Mapped detections to MITRE TTPs targeting MFA exhaustion and social engineering.
- **Advanced Monitoring:** Built alert logic for repeated MFA prompts and abuse patterns across authentication platforms.

📁 GCP THREAT DETECTION DEVELOPMENT

Researched and implemented GCP-native detections across Cloud Storage, Compute Engine, IAM roles, and audit logs—to enhance detection posture.

- **Detection Coverage:** Created rules identifying suspicious role changes, public bucket exposures, and privilege escalation paths.
- **Cross-Team Collaboration:** Worked with GCP security teams to standardize alerts and reduce alert fatigue via contextual tuning.

📁 MICROSOFT DEFENDER FOR CLOUD APPS

Co-led and build detection rules targeting suspicious cloud usage, ransomware indicators, and unauthorized app behavior.

- **Rule Development:** Authored detections across IAM, anomaly-based monitoring, and unknown app discovery.
- **Threat Focus:** Efforts on ransomware detection, suspicious uploads, and credential abuse.

Tech Stack **Summary:** AWS Security Hub, GuardDuty, Google Workspace, GCP, Defender for Cloud Apps, Okta, Duo, MFA, MITRE ATT&CK, SIEM Integration, Python.

Data Sovereignty & Decentralized Identity/PKI: Led strategic R&D projects across self-sovereign identity, AEAD, and cryptographic benchmarking—delivering patentable innovations and providing cryptographic mentorship across engineering teams. Key projects:

- ▶ Data Sovereignty & Decentralized Identity — Project Lead
- ▶ Optimized AEAD for Video Streaming — Lead Architect
- ▶ NIST Lightweight Cryptography for Industrial IoT — Project Owner
- ▶ Key Management System Industry Evaluation — Lead Analyst

DATA SOVEREIGNTY & DECENTRALIZED IDENTITY

Spearheaded a four-year initiative on data provenance, data privacy, and decentralized (self-sovereign) identity/PKI. Integrated secure PKI, device trust, and sovereign identifiers into architecture aligned with regulatory compliance.

- **Trusted Data Flows:** Designed and implemented a data provenance system using IDS-RAM to ensure auditability, compliance, and secure usage control.
- **Decentralized Identity R&D:** Researched and applied W3C Verifiable Credentials (VCs) and Decentralized Identifiers (DIDs) to enable self-sovereign identity systems.
- **PKI Integration & Mentorship:** Provided crypto mentorship on PKI design, device attestation, and secure boot implementation for hardware trust.

OPTIMIZED AEAD FOR VIDEO STREAMING

Led research on secure, high-performance authenticated encryption (AEAD) schemes. Invented new constructions, analyzed relevant cryptographic standards, and filed two patents.

- **Algorithmic Innovation:** Designed new SIV modes with improved performance and nonce misuse resistance over AES-GCM.
- **Standards-Based Design:** Mapped cryptographic strategy to NIST and IETF standards (e.g., AES-GCM-SIV, HMAC, SHA3).
- **Engineering Integration:** Developed PoCs with OpenSSL and guided secure integration into production systems.

NIST LIGHTWEIGHT CRYPTOGRAPHY FOR INDUSTRIAL IOT

Defined and executed an enterprise-wide initiative to evaluate cryptographic algorithms for resource-constrained platforms. Delivered guidance, benchmarking, and integration support.

- **Roadmap & Strategy:** Defined clear project goals and strategy, and gathered performance and security requirements for IoT applications.
- **Algorithm Benchmarking:** Led team evaluating 19+ LWC algorithms (e.g., NIST, CAESAR, eSTRAEM) for seamless implementation.
- **Technical Enablement:** Mentored four engineers and delivered results to executive leadership for adoption planning.

KEY MANAGEMENT SYSTEM INDUSTRY EVALUATION

Analyzed market-leading KMS platforms for enterprise adoption. Benchmarked against NIST and regulatory standards to support secure, scalable deployment strategies.

- **Security Analysis:** Assessed security models, compliance, and scalability of commercial KMS products.
 - **Standards Alignment:** Evaluated designs against NIST KMS best practices and compliance requirements.
- Tech Stack **Summary:** IDS-RAM, W3C DIDs & VCs, AEAD, AES-GCM-SIV, SIV modes, PKI, Secure Boot, Device Attestation, OpenSSL, HSMs, KMS, NIST LWC, CAESAR, IIoT Security, Python, Scripts.
- [Post-Quantum Cryptography Researcher \(PQC\) – ISARA](#) *Jan 2016 – Apr 2019*
- PQC migration Lead & Architect:** Led foundational R&D in PQC readiness, PKI modernization, and secure protocol design. Delivered patents and proprietary IP, defined technical strategy, and contributed to industry standards. Key projects:
- ▶ Quantum-Resilient Automotive PKI — Lead Crypto Engineering
 - ▶ Post-Quantum Cryptography for PKI — Principal Researcher
 - ▶ Post-Quantum TLS 1.3 & PAKE — Protocol Analyst
 - ▶ Vulnerability Assessments & Cryptographic Enablement — Consultant & Mentor
 - ▶ Intellectual Property, Standards Development & Publications — Main Author
-  **QUANTUM-RESILIENT AUTOMOTIVE PKI**
- Led R&D on post-quantum PKI integration for automotive systems under the Security Credential Management System (SCMS). Focused on certificate validation, key lifecycle management (e.g., rotation, recycling), and KEM-based quantum-safe key establishment.
- **Lifecycle Innovation:** Designed public key rotation, validation, and recycling mechanisms optimized for automotive and resource-constrained environments.
 - **KEM Strategy:** Evaluated KEMs and quantum-safe signature schemes as alternatives to traditional DH-based protocols for hybrid and post-quantum key establishment.
-  **POST-QUANTUM CRYPTOGRAPHY FOR PKI**
- Directed R&D on post-quantum cryptographic primitives and migration strategies, with a focus on PKI integration and long-term secure communications.
- **Future-Proof PKI:** Designed hybrid approaches for embedding post-quantum keys and signatures within traditional X.509 certificate infrastructure.
 - **Standards Alignment:** Analyzed NIST PQC candidates and their integration pathways into X.509-based PKI trust chains.
-  **POST-QUANTUM TLS 1.3 & PAKE**
- Researched and scoped quantum-resistant adaptations of TLS 1.3 and password-authenticated key exchange (PAKE) protocols.
- **PQ TLS Integration:** Led engineering efforts to prototype TLS 1.3 with post-quantum key exchange and signature schemes.
 - **PAKE Research:** Defined design and security criteria for post-quantum PAKE protocols aligned with emerging IETF drafts.
-  **SECURITY REVIEWS & ENABLEMENT**
- Advised internal teams on cryptographic implementation, secure engineering, and risk mitigation. Delivered internal training and resolved complex cryptographic challenges.
- **Vulnerability Discovery:** Identified critical flaws in design and implementation of PKI and encryption systems.

- **Team Enablement:** Created crypto tutorials, documentation, and best-practice guidance for engineering teams.

INTELLECTUAL PROPERTY & PUBLICATIONS

Contributed to Compnay's IP portfolio and peer-reviewed cryptographic research, influencing product design and industry standards.

- **Intellectual Property:** Co-inventor on patents related to quantum-safe PKI and post-quantum protocol integration: [3, 4, 5, 7, 6, 8].
- **Publications:** Co-authored multiple papers in cryptography conferences and workshops: [18, 19, 20, 21].
- **Standards Research:** Analyzed and contributed to post-quantum protocols and primitives, including AEAD, HMAC, SHA-2, SHA-3, key agreement, digital signatures, randomized hashing, and PKI.

Tech Stack PQ Transition, Automotive PKI, X.509, Hybrid Certificates, Key Rotation, Digital Signatures & KEMs, Hashing, Merkle trees; PAKE, TLS 1.3, AEAD, Vulnerability Discovery.

Research Assistant – Crypto Group, UCLouvain

Jun 2006 – Jul 2008

Applied Cryptography & Security Researcher: Conducted R&D — including industry collaboration — on cryptographic protocols, RFID system security, and database access-control models with the Crypto Group, Belgium. Focused on practical analysis and implementation aspects. Key projects:

- ▶ Electronic Passports (ePassport) Security Evaluation (RFID) — Crypto Analyst
- ▶ Digital Rights Management for Video Surveillance Systems — Crypto Engineer
- ▶ Stream Cipher Implementation (ECRYPT Project) — Crypto Engineer

DIGITAL RIGHTS MANAGEMENT FOR VIDEO SURVEILLANCE SYSTEMS

Designed and implemented a secure DRM framework incorporating Role-Based Access Control (RBAC). Developed database security modules using PostgreSQL and MySQL, and Java-based middleware for access-control enforcement.

ELECTRONIC PASSPORT (EPASSPORT) SECURITY EVALUATION (RFID)

Performed a comprehensive security assessment of ePassport protocols. Analyzed ICAO and EU standards, identified cryptographic and protocol-level weaknesses using Python, and proposed practical countermeasures.

Financial Database & System Administrator – MEAB

Jul 1997 – Oct 2001

- Managed user accounts, assigned roles, and implemented access control policies.
- Coordinated with head office IT department on system operations.
- Resolved end-user IT issues and provided technical support.
- Developed SQL programs on AS/400 platform to support database tasks.

Postdoctoral Research Fellow – IQC, U. Waterloo/Calgary Jun 2013 – Dec 2015

Conducted advanced research in post-quantum cryptography, under the supervision of David Jao, Michele Mosca, & Rei Safavi-Naini, focusing on the development of quantum-resistant cryptographic primitives and protocols.

- Designed and analyzed digital signatures, and authenticated encryption schemes.
- Supervised and mentored Master's students in cryptographic research methodologies.
- Publications: [12, 22, 17].

Education

Jan 2009 – **Ph.D. (Quantum Computer Science, Cryptography, Information Security)**, *Computer Science and Operations Research, LITQ, Université de Montréal, Canada*
May 2013

Title Key Agreement Against Quantum Adversaries.

Supervisors Gilles Brassard and Louis Salvail.

Research My research topic dealt with key agreement protocols over classical communication channels against quantum adversaries. Part of my achievements:

- Providing the *first* classical protocol secure against quantum adversaries in the random oracle model, which was conjectured to be *impossible* to exist.
- Providing better sequences of classical and quantum protocols.
- Results were presented at Crypto'11. Further results were presented at QCrypt'11 and QIP'12. These are top-tier conferences.
- Publications: [9, 13, 14].
- Skills: Information Security, Quantum-Safe Crypto, Quantum Algorithms, PKI, Quantum Key, Cryptanalysis, Grover-based algorithms.

Jan 2002 – **M.Sc. (Computer Science, Efficient Algorithms for Cryptography)**, *Computer Science and Operations Research, LITQ, Université de Montréal, Montreal, Canada*
May 2006

Title Implementation (C++ and HW) of Large Number Multiplication by FFT in the Context of Cryptographic Algorithms.

Supervisors Jean-Pierre David and Gilles Brassard

Research & Development

- Presenting a thorough study of the most efficient multiplication algorithms and their FPGA implementations in the context of cryptosystems.
- Implementation in SW/HW a new design of the Fast Fourier Transform (FFT) in modular arithmetic to efficiently compute large integers multiplication.
- Publications: [10, 16].

Publications

Patents

- [1] Kassem Kalach, Ali Abbasi, and Teng Wu. **New Efficient SIV Design**. Patent filed, October, 2020.
- [2] Kassem Kalach, Ali Abbasi, and Teng Wu. **Poly1305-SIV: Stronger, Faster than AES-GCM**. Patent to be filed, February, 2021.
- [3] Kassem Kalach. **Refreshing public parameters in lattice-based cryptographic protocols**. Patent number: 9673977, Jun 2017. I am using a more descriptive title "Defeating all-for-the-price-of-one and backdoor attacks in lattice cryptography".
- [4] Kassem Kalach and Anton Mosunov. **Pairing verification in supersingular isogeny-based**

cryptographic protocols. Patent number: 10116443, October 30, 2018.

- [5] Kassem Kalach and Anton Mosunov. **Public key validation in supersingular isogeny-based cryptographic protocols.** Patent number: 10313124, Jun 04, 2019.
- [6] Shane Daniel Kelly, Filip Kostis Pawlega, and Kassem Kalach. **Generating a lattice basis for lattice-based cryptography.** Patent number: 10097351, October 9, 2018.
- [7] Victoria de Quehen, Kassem Kalach, and Anton Mosunov. **Supersingular Isogeny Key Encapsulation Based on Twisted Edwards Curves.** Patent filed, Dec 5, 2018.
- [8] Edward Eaton, Atsushi Yamada, and Kassem Kalach. **Using conversion schemes in public key cryptosystems.** Patent number: 10031795, July 24 2018.

Journal Articles

- [9] Gilles Brassard, Peter Høyer, Kassem Kalach, Marc Kaplan, Sophie Laplante, and Louis Salvail. **Key Establishment à la Merkle in a Quantum World.** *Journal of Cryptology*, April 2019.
- [10] Jean-Pierre David, Kassem Kalach, and Nicolas Tittley. **Hardware Complexity of Modular Multiplication and Exponentiation.** *IEEE Transactions on Computers*, 56(10):1308–1319, October 2007.

Conference Proceedings

- [11] Gildas Avoine, Kassem Kalach, and Jean-Jacques Quisquater. **ePassport: Securing International Contacts with Contactless Chips.** In *Financial Cryptography and Data Security, 12th International Conference, FC 2008, Mexico, January 28-31*, volume 5143 of LNCS, pages 141–155, 2008.
- [12] Reza Azarderakhsh, David Jao, Kassem Kalach, Brian Koziel, and Christopher Leonardi. **Key Compression for Isogeny-Based Cryptosystems.** In *Proceedings of the 3rd ACM International Workshop on ASIA Public-Key Cryptography, AsiaPKC, Xi'an, China, May 30 - June 03, 2016*, pages 1–10, 2016.
- [13] Gilles Brassard, Peter Høyer, Kassem Kalach, Marc Kaplan, Sophie Laplante, and Louis Salvail. **Merkle Puzzles in a Quantum World.** In *QCRYPT 2011: First Annual Conference on Quantum Cryptography*, ETH, Switzerland, September 12, 2011.
- [14] Gilles Brassard, Peter Høyer, Kassem Kalach, Marc Kaplan, Sophie Laplante, and Louis Salvail. **Merkle Puzzles in a Quantum World.** In Phillip Rogaway, editor, *Advances in Cryptology - Proceedings of CRYPTO 2011 - 31st Annual Cryptology Conference, Santa Barbara, CA, USA, August 14-18*, volume 6841 of LNCS, pages 391–410, 2011. The complete version can be found at <http://arxiv.org/abs/1108.2316>.
- [15] Philippe Bulens, Kassem Kalach, François-Xavier Standaert, and Jean-Jacques Quisquater. **FPGA Implementations of eSTREAM Phase-2 Focus Candidates with Hardware Profile.** In *The State of the Art of Stream Cipher (SASC)*, Germany, Jan 2007.
- [16] Kassem Kalach and Jean-Pierre David. **Hardware implementation of large number multiplication by FFT with modular arithmetic.** In *The 3rd International IEEE-NEWCAS Conference, 2005*, pages 267–270, Canada, June 2005.
- [17] Kassem Kalach and Reihaneh Safavi-Naini. **An Efficient Post-Quantum One-Time Signature Scheme.** In *Selected Areas in Cryptography - SAC - 22nd International*

Conference, Sackville, NB, Canada, August 12-14, volume 9566 of *LCNS*, pages 331–351, 2015.

Technical Report

- [18] Paul Bottinelli, Kassem Kalach, Filip Pawlega, and Atsushi Yamada. **Towards Quantum-Resistant Vehicular PKI**. Technical report, ISARA Corporation, Dec 21, 2018.
- [19] Edward Eaton, Kassem Kalach, Philip Lafrance, Alex Parent, and Atsushi Yamada. **QC-MDPC KEM: A Key Encapsulation Mechanism Based on the QC-MDPC McEliece Encryption Scheme**. Technical report, Submission to NIST's Post-Quantum Cryptography Standardization Project, ISARA Corporation, 2017. Available at <https://csrc.nist.gov/projects/post-quantum-cryptography/round-1-submissions>.
- [20] Kassem Kalach. **Security Assessment of HIMMO: A Lightweight Collusion-Resistant Key Predistribution Scheme**. ISARA Corporation, February, 2016.
- [21] Kassem Kalach, Alex Parent, Jerry Sui, and Atsushi Yamada. **QC-MDPC McEliece Quantum Security Analysis**. Technical report, ISARA Corporation, July 2017.

Unpublished

- [22] David Jao and Kassem Kalach. **One-Time Signatures Using Generalized Discrete Logarithm Problem**. Theoretical construction is completed, an instantiation is under study, IQC, University of Waterloo, 2017.
- [23] Kassem Kalach, Alfred Menezes, and Michele Mosca. **Efficient Authenticated Encryption Using Quantum Key Distribution (QKD)**. IQC, University of Waterloo.

Selected Talks

International/External Talks

- December 12–13, 2017 **Isogeny-Based Cryptography**, *CSE Research Workshop*, Communications Security Establishment (CSE), Ottawa, Canada
- August 14, 2015 **Efficient One-Time Digital Signatures**, *22nd International Conference on Selected Areas in Cryptography (SAC 2015)*, Sackville, Canada
- Jun 21, 2012 **Key Establishment à la Merkle in a Quantum World**, *CSCQI 2012: The 9th Canadian Student Conference on Quantum Information*, Institute for Quantum Computing, University of Waterloo, Canada
- September 12, 2011 **Merkle Puzzles in a Quantum World**, *QCRYPT 2011: First Annual Conference on Quantum Cryptography*, ETH Zurich, Switzerland
- August 17, 2011 **Merkle Puzzles in a Quantum World**, *CRYPTO 2011: The 31st International Cryptology Conference*, University of California, Santa Barbara, USA

Selected Internal Seminars/Tutorials

- April 2024– **Understanding PKI series**, *UKG Seminars*, Canada
- May 2019– **All Projects Cited Above and Beyond (AEAD, LWC, KMS, Machine Learning Using ZKP, VCs, DID, etc.)**, *100s of Slides & 10s of Talks*, Huawei, Canada
- TLS 1.2 vs TLS 1.3**, *ISARA Seminars*, Canada
- Password-based AKE Protocols**, *ISARA Seminars*, Canada
- Authenticated Key Exchange (AKE) for TLS 1.3**, *ISARA Seminars*, Canada
- Polynomials multiplication using Fast Fourier Transform (FFT)**, *ISARA Seminars*, Canada

Security Assessment of HIMMO: A Lightweight Collusion-Resistant Key Predistribution Scheme, ISARA Seminars, Canada

Awards, Scholarships, Others

- 2013 Excellent Ph.D. Thesis.
- 2013 Best Thesis Candidate, Computer Science Department.
- 2011 Crypto'11 Best Paper Candidate.
- 2013 Excellent M.Sc. Thesis.

Languages

English, Fr	Full professional proficiency	<i>comprehension, writing, speaking</i>
German	Basic	<i>comprehension, writing, speaking</i>
Arabic	Native proficiency	<i>comprehension, writing, speaking</i>

Selected International Conferences

- May 21–23, 2024 Cloud and Cybersecurity Expo, *Messe Frankfurt*, Frankfurt, Germany.
- October 16–18, 2018 The 25th ACM Conference on Computer and Communications Security (CCS'18), *ACM*, Toronto, Canada.
- April 9–11, 2018 The Ninth International Conference on Post-Quantum Cryptography, PQCrypto 2018, *University of Florida, Fort Lauderdale, Florida*, United States.
- August 16–18, 2017 The 24th Annual Conference on Selected Areas in Cryptography (SAC'17), *University of Ottawa*, Ottawa, Canada.
- Jun 12,19,26, 2015 Workshop on Commercialization and Entrepreneurship, *CryptoWorks21*, *University of Waterloo*, Canada.
- April 2–3, 2015 Workshop on Cybersecurity in a Post-Quantum World, *Information Technology Laboratory, NIST*, USA.
- October 1–3, 2014 PQCrypto 2014, 6th International Conference on Post-Quantum Cryptography, *CryptoWorks21*, *University of Waterloo*, Canada.
- November 26–28, 2013 Fundamentals of network security, *CryptoWorks21*, *University of Waterloo*, Canada.
- July 23–26, 2013 Workshop on Project Management, Standardization & Certification, *CryptoWorks21*, *University of Waterloo*, Canada.
- June 25–28, 2013 International Conference on Applied Cryptography and Network Security (ACNS'13), *ISPIA*, Banff, Canada.
- September 12–16, 2011 QCRYPT 2011: First Annual Conference on Quantum Cryptography, *ETH Zurich*, Switzerland.
- August 14–18, 2011 CRYPTO 2011: 31st Annual International Conference on Cryptology, *University of California*, Santa Barbara, USA.
- May 5, 2006 Workshop on Computer Privacy in Electronic Commerce, *Université de Montréal*, Canada.
- May 2–6, 2004 EUROCRYPT 2004, *Network Security and Cryptography Group at the IBM Zurich Research Laboratory & IACR*, Interlaken, Switzerland.